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High-Frequency Evidence on Post-Covid Treasury Yields and Bond-Stock Comovement

Prepared by Jing Zhou*

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ABSTRACT: Since mid-2021, long-term Treasury yields have risen by more than 300 basis points, and Treasuries have provided less offset to stock market losses. We study both changes in a single high-frequency framework that uniformly measures yield moves in narrow windows around major macro data releases, Treasury debt issuance announcements and auctions, and Federal Reserve monetary policy announcements, and we analyze the role of all three jointly. Before Covid, most of the movement in yields and of their comovement with stocks occur outside these windows. After Covid, both move into macro and debt issuance windows. Together, these windows occupy 4 percent of trading time but account for over 60 percent of the rise in the 10-year yield and most of the positive bond-stock comovement, more so at longer maturities.

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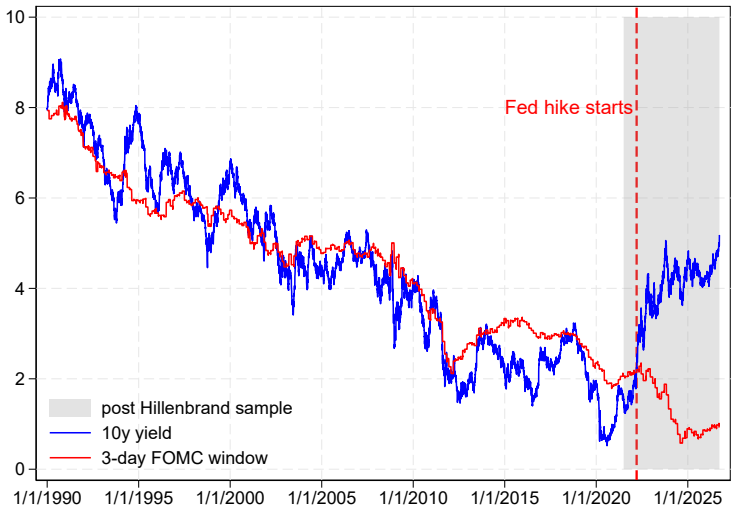
High-Frequency Evidence on Post-Covid Treasury Yields and Bond-Stock Comovement

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1 Introduction

Long-term U.S. Treasury yields have risen sharply since Covid, with the 10-year yield closing above 5 percent in September 2026 for the first time since 2007. Several forces could be behind the rise, from monetary tightening in response to the post-Covid inflation to heavier Treasury issuance on the back of large fiscal deficits. Where the rise was priced stands in contrast with the preceding decline. Using data through mid-2021, Hillenbrand (2025) documents that essentially the entire decline in the 10-year yield since 1990 occurred in three-day windows around the Federal Reserve monetary policy announcements (FOMC meetings), while yield movements outside those windows largely canceled out.

Figure 1: 10-Year U.S. Treasury Yield and Cumulative Change Around FOMC



Note: This figure plots the daily 10-year U.S. Treasury yield and its cumulative change in 3-day windows around FOMC meetings.

This pattern breaks down after mid-2021. Between July 2021 and August 2026, the 10-year yield rose by over 300 basis points (bp), while its cumulative change in three-day FOMC windows was −108bp (Figure 1). The decline in long rates was priced around FOMC meetings, but the recent rise was not. This raises two questions: where were these changes priced, and what does that reveal about their drivers?

To answer these, we extend the time-stamped event data of Kerssenfischer and Schmeling (2024) and study three categories of events from January 2013 through August 2026: (i)

macro events, 18 major data releases about growth and inflation; (ii) *Treasury debt issuance events*, namely quarterly refunding announcements, note and bond auction announcements, which carry news about Treasuries supply (Phillot, 2025), and auction results, which carry news about Treasuries demand (Ray, Droste and Gorodnichenko, 2024); and (iii) *FOMC events*, the Federal Reserve's monetary policy statements and press conferences. For every event, we measure the shock in a unified way: the change in yields in a 45-minute window around it.[1] A common measure is key to our analysis. It makes the three categories directly comparable, and because every 15-minute interval lies either in one category of windows or outside all of them, it assigns every intraday yield move to one category or to non-event. Studied one at a time, any single category would leave the rest of the movement in yields unattributed. We use this framework to examine two dimensions of the Treasury market across segments of the yield curve: movements in yields (*first moment*) and their comovement with stocks, the bond-stock beta (*second moment*).

Turning to the first moment, after mid-2021, most of the rise in long yields occurred in macro and debt issuance windows. These windows occupy only 4 percent of trading time yet account for more than 60 percent of the over 300bp increase in the 10-year yield: about half comes in macro windows and about one tenth in debt issuance windows, while FOMC windows lowered the yield slightly. Because an accounting up to a date depends on the end day chosen, we also take a weighted average of each category's contribution share over all days of the sample. By this measure, macro and debt issuance windows account for about 70 percent of the 10-year yield movement after Covid, whereas before Covid non-event accounts for most of it. The pre-Covid pattern does not contradict Hillenbrand (2025), whose result spans three decades. By the same measure, his three-day FOMC windows account for 95 percent of the path of the 10-year yield over 1990–mid-2021, but 48 percent over 2013–mid-2021, i.e., our pre-Covid sample. Moreover, his by-day decomposition shows that the days before and after an FOMC meeting, which lie outside our 45-minute windows, account for more than half of the three-day move.

The contributions from macro and debt issuance windows are not transitory. Following

[1]The window spans $[-15, +30)$ minutes, slightly wider than the usual $[-10, +20)$ minutes in the monetary policy literature. Most events occur on the hour, quarter past, or half hour, so a 15-minute grid places every interval cleanly inside or outside an event window. Furthermore, a 45-minute window remains narrow enough to identify shocks cleanly, while avoiding the fivefold increase in sample size and computational burden from a 5-minute grid.

Altavilla, Giannone and Modugno (2017), we regress yield changes over horizons of up to a quarter on the sum of shocks over the same horizon, and find pass-through coefficients close to one: shocks are neither reversed nor amplified. Pass-through is similar before and after Covid, so the larger post-Covid contributions reflect larger average shocks, not a change in how they propagate.

Within macro windows, the larger average shock comes from real-activity releases rather than prices: e.g., the 10-year yield rises by about 110bp in employment report windows alone. Bloomberg consensus forecast errors show that this is not because headline forecast errors widened after 2021, or yields became more sensitive to them. Sensitivity rose only for price releases, mainly during the 2022–24 tightening. Headline forecast errors account for about one tenth of the macro contribution, and the remainder is consistent with news that headline forecast errors miss, such as revisions and underlying components of releases beyond headline (Gürkaynak, Kısacıkoğlu and Wright, 2020).

The two categories move different components of yields. In local projections on the expected rate and term premium, debt issuance shocks raise the 10-year yield almost entirely through term premium, whereas macro and FOMC shocks move it mainly through expected short rates. The expected-rate response to macro shocks persists for at least three months, while their term-premium response fades within a month. The role of macro and debt issuance windows also rises with maturity: they account for a quarter of the rise in the 2-year yield, half for the 2–5-year forward, and 80 to 90 percent for the 5–10 and 10–30-year forwards.

Turning to the second moment, the same framework locates where Treasuries become a poor hedge for equities. Measured at 15-minute frequency, the bond-stock beta flips from negative to positive between the two samples, at every segment of the yield curve. Because every 15-minute interval belongs to one event category or to non-event, the beta decomposes exactly into contributions from each. Before Covid, non-event accounts for over 90 percent of the negative beta at long-maturity forward. After Covid, the positive beta arises almost entirely inside windows: macro accounts for about 70 percent and debt issuance for about 15 percent, while non-event contributes about 10 percent. Furthermore, the flip comes from how bonds and stocks comove within windows, not from windows carrying more equity market variance, as the variance shares remain somewhat stable between periods. Along the curve, FOMC windows matter most at the front end, while the debt issuance contribution rises with maturity, to about a quarter at the 10–30-year forward.

Taken together, the two moments point to a clear pattern. Before Covid, most of the movement in long yields and of their comovement with stocks occurred outside scheduled windows. After Covid, both move into macro and debt issuance windows. The evidence is consistent with two channels operating in these windows. In macro windows, data releases raise expected policy rates and lower stock prices, as markets price the Fed's reaction to the data. In debt issuance windows, news about Treasury supply and demand raises the term premium and lowers stock prices, consistent with a rise in risk premia common to bonds and stocks that is stronger at longer maturities.

Related literature. This paper contributes to three strands of literature. First, it relates to high-frequency studies of how news moves Treasury yields. Most of this work examines one type of news at a time: FOMC announcements (e.g., Kuttner, 2001; Nakamura and Steinsson, 2018), macro releases (e.g., Gilbert et al., 2017; Altavilla, Giannone and Modugno, 2017), or Treasury supply and demand (e.g., Ray, Droste and Gorodnichenko, 2024; Phillot, 2025; Bi, Phillot and Zubairy, 2026). Kerssenfischer and Schmeling (2024) pool these event types, but study their contribution to the variance of yields through 2020. We measure all three in the same way and analyze both the changes in yields and their comovement with stocks to them over the post-Covid episode.

Second, it relates to work on the drivers of long-term rates. The decline in long rates before Covid has been attributed to secular stagnation (e.g., Rachel and Summers, 2019) and to Fed communication (Hillenbrand, 2025). For the post-Covid reversal, the literature points to a shift in the marginal buyers of Treasuries (Chaudhary, Fu and Zhou, 2024), a rising term premium amid heavier coupon issuance (Diercks and Asnani, 2024), and a possibly higher neutral rate (Rachel, 2025). Two contemporaneous papers close to ours, Christensen and Rudebusch (2026) find that fiscal, AI, and FOMC news in three-day windows does not explain the rise in r^*, and Beaudry, Cavallino and Willems (2026) attribute most of the rise in yields to employment-report and prominent Fed-speech days. We depart from these studies in three ways: we identify shocks in 45-minute windows rather than days, trace the impact of each shock to expected rate and term premium, and show that the windows that raise yields are also those where Treasuries turn to a poor hedge for equities.

Third, it relates to work on bond-stock beta. A positive beta has been linked to the covariance of inflation and growth (Campbell, Pflueger and Viceira, 2020), to supply-driven

inflation under an inflation-focused Fed (Pflueger, 2025), and to the interaction of monetary-fiscal regimes with the mix of macroeconomic shocks (Li et al., 2022). Closest to us, Hu, Jin and Pan (2025) construct a daily stock-bond correlation from intraday data, show that Treasuries are disproportionately risky on FOMC days, and document its rise after 2021. Decomposing the beta by event categories beyond FOMC, we show that the post-Covid flip arises within macro and Treasury debt issuance windows, while comovement outside them vanishes. This is consistent with the mechanism of Pflueger (2025) and adds a fiscal channel at the long end.

The remainder of the paper is organized as follows. section 2 describes the data and the event classification. section 3 presents the first-moment results: the accounting, the persistence of window moves, their sources, the split between expected rates and the term premium, and the pattern across the curve. section 4 turns to the bond-stock beta and its decomposition by event category and across the curve. section 5 concludes.

2 Data and Event Classification

2.1 High-Frequency Asset Prices

Our asset price data consist of intraday quotes for U.S. Treasury securities and equities from Bloomberg tick-level data, aggregated to 15-minute frequency over January 2013 through August 2026. For Treasuries, we use nominal yields at the 2-, 5-, 10-, and 30-year maturities, from which we form four curve segments: the 2-year spot rate and the forward rates $f(2,5)$, $f(5,10)$, and $f(10,30)$. Working with forwards rather than spot yields isolates the pricing of distinct portions of the curve: the $f(10,30)$ segment, for instance, reflects the pricing of duration far beyond the horizon over which the current policy cycle is expected to play out.

For equities, we use the front-month E-mini S&P 500 futures contract, which trades nearly continuously and therefore remains liquid during the pre-open releases that account for a substantial share of our macro events. We restrict the sample to regular trading hours and drop the 15-minute intervals affected by holidays and early closes.

2.2 Event Time Stamps

Our event calendar extends the time-stamped database of Kerssenfischer and Schmeling (2024), which records the precise release time of scheduled macroeconomic, Treasury debt issuance, and monetary policy events. We augment it in two ways: we extend the sample through August 2026, and we add the Treasury's quarterly refunding announcements, which are absent from the original database but are a key debt issuance event.

Macro events. We restrict to 18 key releases that are also covered by Bloomberg's economic forecast data. The set spans real activity (employment report, jobless claims, retail sales, industrial production, durable goods orders, GDP, housing starts and permits, new home sales, trade balance, leading index), prices (CPI, PPI, unit labor costs and productivity, employment cost index), and surveys (ISM manufacturing, ISM non-manufacturing, University of Michigan consumer sentiment, Conference Board consumer confidence). For each release, we pair the realized value with the Bloomberg consensus forecast recorded immediately before, which yields the forecast errors used in subsection 3.4. We scale each release's forecast errors by their standard deviation over January 2013-February 2020 and winsorize them at ± 5. The same scaling enables a consistent comparison across time, and using pre Feb 2020 drops pandemic outliers, such as the June 2020 payroll report.

Debt issuance events. We include three types of Treasury announcements. First, the quarterly refunding announcement (QRA), in which the Treasury announces its financing estimates and the size and composition of coupon issuance for the coming quarter. Second, nominal note and bond auction announcements, which specify the size and maturity of an upcoming auction and therefore convey news about the supply of Treasuries to be absorbed (debt supply shocks, Phillot, 2025). Third, nominal note and bond auction results, released at the auction close, which reveal the bid-to-cover ratio, the tail relative to the when-issued yield, and the allocation across bidder types, and therefore convey news about the demand for Treasuries (debt demand shocks, Ray, Droste and Gorodnichenko, 2024). We restrict the auction set to nominal coupon securities, excluding bills, TIPS, and FRNs, so that the events map onto the maturities in our asset prices data.

FOMC events. We include the release of the FOMC statement and the press conference as separate events, since statements and press conferences represent distinct phases of FOMC communication and may convey different information (Cieslak, McMahon and Pang, 2024).

When events of different categories fall in overlapping windows (63 of about 4,000 event windows), we assign the window by the priority FOMC > debt issuance > macro. Within the macro category, the priority is by the share of Bloomberg users who set an alert for the release, as reported by Altavilla, Giannone and Modugno (2017). We split the sample at 30 June 2021, yielding a pre-period of January 2013–June 2021 and a post-period of July 2021–August 2026. The break date is chosen for two reasons. It coincides with the end of the sample in Hillenbrand (2025), which makes our pre-period directly comparable to the regime he documents. It also precedes by several months the inflation surprises of late 2021 and the subsequent tightening cycle, so the post-period contains the full monetary tightening episode rather than beginning in its midst.

2.3 Event-Window Shocks

For each event j occurring at time τ_j, we define the shock as the change in the asset price over a 45-minute window, following Kerssenfischer and Schmeling (2024),

$$s_j \;=\; i_{\tau_j+30\mathrm{m}} \;-\; i_{\tau_j-15\mathrm{m}}, \tag{1}$$

where i denotes the yield on a given curve segment or the log equity price. Each window thus spans three 15-minute intervals. For FOMC press conferences, the window extends to 60 minutes after the start. The 15-minute pre-event leg allows for pre-positioning and for small imprecision in recorded release times; the 30-minute post-event leg is long enough for the information to be absorbed but short enough that unrelated news is unlikely to contaminate the window. We refer to the 45-minute yield change as the macro (debt issuance, or FOMC) *shock*, and to the Bloomberg forecast error as macro *news* or *surprise*.

3 First Moment Results: Change in Yields

This section examines the role of macro, debt issuance, and FOMC shocks in the post-Covid rise of long-term yields. We begin with an accounting that assigns every 15-minute change

in yields to the event category whose window contains it (subsection 3.1). We then ask whether window moves persist, using the low-frequency regressions of Altavilla, Giannone and Modugno (2017) (subsection 3.2). Because pass-through is stable across the two samples, the larger post-Covid role of events must come from the average move per event, which we examine next (subsection 3.3), before asking whether larger macro moves reflect larger news or a stronger response to it (subsection 3.4). We then split the effect of each category into expected short rates and the term premium (subsection 3.5), and close with the pattern across the yield curve (subsection 3.6).

3.1 Accounting

Because every 15-minute interval is assigned to exactly one event category or to non-event (section 2), the change in the yield over any sample splits exactly into contributions from each category. Let Δi_t denote the 15-minute change in the 10-year yield in interval t, and let $\mathcal{B}^k$ denote the set of intervals in the windows of category $k \in \{m, d, f, n\}$ (macro, debt issuance, FOMC, and non-event). The total yield change splits into four additive pieces,

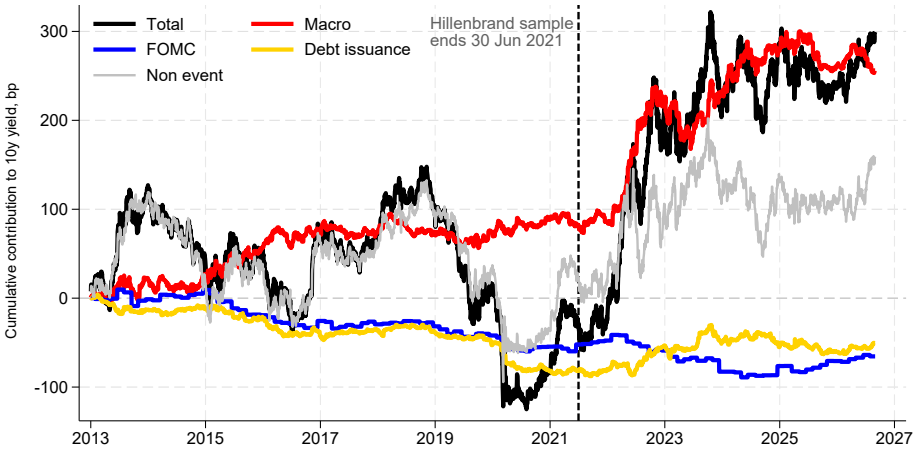
$$i_T - i_0 = \sum \Delta i_t = \underbrace{\sum_{t \in \mathcal{B}^m} \Delta i_t}_{C^m} + \underbrace{\sum_{t \in \mathcal{B}^d} \Delta i_t}_{C^d} + \underbrace{\sum_{t \in \mathcal{B}^f} \Delta i_t}_{C^f} + \underbrace{\sum_{t \in \mathcal{B}^n} \Delta i_t}_{C^n}. \tag{2}$$

The decomposition is an identity and requires no model. Its content lies in how each contribution C^k compares with the share of trading time that category k occupies: if event windows carried no more information than ordinary trading, each category would contribute roughly in proportion to its time share.

The cumulative contributions reveal two distinct regimes on either side of mid-2021 (Figure 2). After mid-2021, macro and debt issuance windows account for the majority of the rise in the 10-year yield while occupying only about 4 percent of trading time (Table 1). Macro windows occupy 2.6 percent of trading time but account for 174bp (53 percent) of the 331bp rise. Per unit of time, they moved the yield about twenty times as much as an average interval. Debt issuance windows add another 31bp (9 percent) in 1.4 percent of trading time. Together, macro and debt issuance windows account for about 60 percent of the rise, while FOMC windows subtract 13bp.

The pre-Covid sample shows a different pattern, with event categories pulling in opposite

Figure 2: 10-Year U.S. Treasury Yield and Cumulative Change By Event Category



Note: This figure plots the 10-year yield and its cumulative change from macro, debt issuance, or FOMC event windows, at 15-minute frequency. The yield is normalized to zero at the start of the sample.

directions and none driving the yield. The 10-year yield fell by 32bp; FOMC windows contributed −52bp, in line with Hillenbrand (2025), and debt issuance windows −82bp. The debt issuance decline comes from note auction results, after which the yield fell by about 0.2-0.3bp on average (Table A4). Macro windows, by contrast, added 82bp, against the direction of the overall decline.

Table 1: Contribution of 10-Year Yield Changes By Event Category

	Pre: 2013–Jun 2021				Post: Jul 2021–Aug 2026			
	Δ10y (bp)	% of Δ10y	% trading hour	N of events	Δ10y (bp)	% of Δ10y	% trading hour	N of events
FOMC	-52.0	164.4	0.2	117	-13.4	-4.1	0.2	82
Debt issuance	-81.7	258.1	1.3	792	31.0	9.4	1.4	518
Macro	82.1	-259.5	2.6	1568	174.1	52.6	2.6	965
Residual	19.9	-63.0	96.0	–	139.3	42.1	95.8	–
Total	-31.6	100.0	100.0	2477	330.9	100.0	100.0	1565

Note: This table reports the cumulative 10-year yield changes in the macro, debt issuance, and FOMC event windows and in the non-event intervals, in basis points, contribution (in percent) to total 10-year yield change, together with each category's share of 15-minute trading intervals and number of events, for the pre and post samples, respectively.

The decomposition in Equation 2 depends on the end day chosen. Following the logic of integral multipliers (Ramey and Zubairy, 2018), which summarize a dynamic effect over a

whole horizon rather than at a single end point, we summarize each category's contribution over the whole path of the yield. Let $s_\tau^k = C_\tau^k/(i_\tau - i_0)$ denote the share of category k in the cumulative change in the yield up to day τ, the analogue of $C^k/(i_T - i_0)$ in Equation 2. We average these end-day shares over all days, weighting each end day by the distance of the yield from its starting level:

$$\Lambda^k = \sum_\tau w_\tau \, s_\tau^k = \sum_\tau \frac{|i_\tau - i_0|}{\sum_\sigma |i_\sigma - i_0|} \cdot \frac{C_\tau^k}{i_\tau - i_0} = \frac{\sum_\tau \operatorname{sign}(i_\tau - i_0)\, C_\tau^k}{\sum_\tau |i_\tau - i_0|} \tag{3}$$

Since $\sum_k C_\tau^k = i_\tau - i_0$ on every end day τ, the shares Λ^k sum to one. Intuitively, Λ^k measures how much of the area between the yield and its starting level in Figure 2 is covered by the path of category k. The denominator is the area between the yield's path and its starting level. The numerator is the corresponding area for the cumulative contribution of category k, counted as positive when the contribution lies on the same side of the starting level as the yield and as negative if not.

Table 2: Contribution of 10-Year Yield Changes By Event Category, Weighted

	Macro	Debt issuance	FOMC	Non-event	Sum
Pre: 2013–Jun 2021	29.0	-5.7	-1.5	78.2	100.0
Post: Jul 2021–Aug 2026	61.1	10.5	-7.2	35.6	100.0

Note: This table reports Λ^k from Equation 3, in percent: the average over end days of category k's share of the cumulative change in the 10-year yield since the start of the sample, weighting each end day by the absolute size of that change. Cumulative changes are measured at the end of each trading day. Shares sum to 100 by construction.

Like Table 1, the weighted contributions show that macro and debt issuance windows became the main contributors to the 10-year yield after Covid (Table 2). Before mid-2021, non-event intervals account for 78 percent of the path of the 10-year yield, and macro and debt issuance windows together for less than a quarter. After mid-2021, the pattern reverses: macro windows account for 61 percent of the path and debt issuance windows for 11 percent, while the non-event share falls to 36 percent. These post-Covid shares are similar to those in Table 1, 72 against 62 percent for macro and debt issuance windows combined, because the yield rises fairly monotonically over the period, so the choice of end day matters little. Before Covid, by contrast, the yield moved above and below its starting level and ended the sample close to it, so the shares in Table 1 depend heavily on the end day, and the weighted shares show that non-event trading accounted for most of the yield's movement, averaged

across the period.

Beyond the three event categories, the growing issuance of long-dated bonds by AI-related companies has been cited as another driver of Treasury yields.[2] The argument is that heavy AI-related corporate bond supply competes with Treasuries for the same investors and pushes Treasury yields up. We find no significant movement in yields around AI-related corporate bond announcements: across the 15 deals since January 2025, the 10-year yield moves by an insignificant 0.25bp per announcement on average (Appendix A).

3.2 Pass-through of Event Shocks to Yields

High-frequency windows identify the contemporaneous impact of event but not its persistence. Window moves could partially reverse, for instance if they reflect temporary price pressure or overreaction, in which case the accounting overstates the role of events. Alternatively, information released in an event may continue to be absorbed after the window closes, as documented for FOMC announcements by Brooks, Katz and Lustig (2018), in which case the accounting understates it.

To allow delayed impact, we regress low-frequency yield changes on the sum of window moves over the same interval, following Altavilla, Giannone and Modugno (2017) :

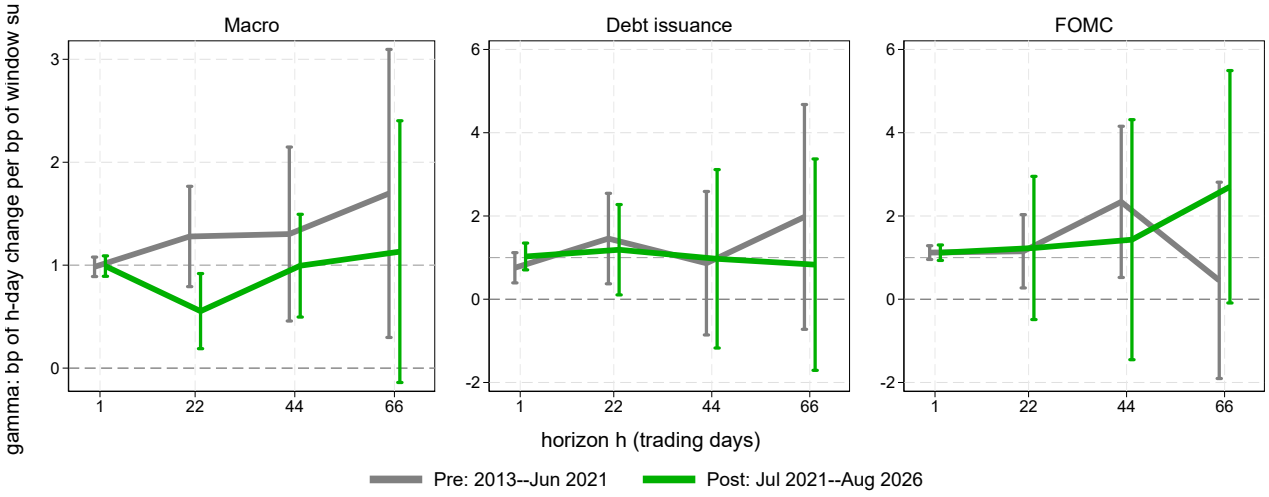
$$\Delta i_{t,t+h} = \underbrace{\gamma_m^h m_{t,t+h}}_{\text{macro}} + \underbrace{\gamma_d^h d_{t,t+h}}_{\text{debt issuance}} + \underbrace{\gamma_f^h f_{t,t+h}}_{\text{FOMC}} + \epsilon_{t+h}, \quad h = 1, 22, 44, 66 \text{ days}, \tag{4}$$

where $\Delta i_{t,t+h}$ is the change in the 10-year yield over h trading days, and $m_{t,t+h}$, $d_{t,t+h}$, and $f_{t,t+h}$ are the sums of the 45-minute window moves of macro, debt issuance, and FOMC events within the same h-day span. We estimate Equation 4 on non-overlapping blocks of h days (from 3,444 daily blocks to 53 blocks at $h = 66$ days, roughly a quarter), separately for the pre- and post-Covid samples. A pass-through $\gamma_k^h = 1$ means that window moves persist one for one at horizon h, $\gamma_k^h < 1$ indicates partial reversal, and $\gamma_k^h > 1$ amplification. Our regressors differ from those of Altavilla, Giannone and Modugno (2017), who use Bloomberg forecast errors. The window move summarizes a broader set of information that market prices in, including components and revisions that the headline surprise misses (Gürkaynak,

[2] See, e.g., Bloomberg, "AI Debt Sales Are Keeping US Treasury Yields High", 17 August 2026; Financial Times, "Are Big Tech bonds crowding out the US Treasury?", 24 September 2026.

Kısacıkoğlu and Wright, 2020). It also measures macro, debt issuance, and FOMC events in a common way, overcoming the lack of consensus forecasts covering debt issuance events.

Figure 3: Pass-through Of Event Shocks By Horizon



Note: This figure plots the pass-through coefficients γ_m^h, γ_d^h, and γ_f^h for macro, debt issuance, and FOMC shocks, estimated from Equation 4, respectively. The horizontal axis denotes the horizon of $h = 1,\ 22,\ 44,\ 66$ days. The whiskers denote 90% confidence intervals, based on heteroskedasticity-robust standard errors.

Window moves persist roughly one for one, and pass-through is similar before and after Covid (Figure 3). At $h = 1$, all coefficients are close to one in both samples: window moves are not undone within the day. At longer horizons the point estimates mostly remain near one, with confidence intervals that widen as the number of blocks shrinks. The pre- and post-Covid confidence intervals overlap at every horizon and for every event category, so pass-through has not changed in a statistically significant way. This persistence matches Altavilla, Giannone and Modugno (2017), who also find that the high-frequency effects of macroeconomic news on yields do not revert at lower frequencies.

Allowing for delayed effects leaves the accounting largely unchanged (Table 3). Because pass-through is stable across samples, we estimate it on the full sample and multiply each $\hat{\gamma}_k^h$ by the post-Covid sum of window moves of category k. The macro share is 52 percent at $h = 1$ and 39–58 percent at one month to one quarter, against 53 percent in the raw accounting. The debt issuance share ranges from 8 to 11 percent, against 9 percent raw. The FOMC contribution stays small and negative throughout. At the quarterly horizon, macro and debt issuance events together account for about two-thirds of the 10-year yield rise.

Table 3: Contribution of 10-Year Yield Changes By Event Category, With Longer Horizon Pass-through

	Raw 45-min	Aggregation horizon h (trading days)			
	window sum	$h = 1$	$h = 22$	$h = 44$	$h = 66$
Share of the +330.9 bp post-2021 rise in the 10-year yield (%)					
Macro	52.6	52.1***	39.0***	58.4***	56.1**
Debt issuance	9.4	8.4***	10.8**	8.0	11.1
FOMC	-4.1	-4.6***	-5.9***	-8.0**	-6.1
Blocks	—	3444	157	79	53
R^2	—	0.22	0.21	0.29	0.30

Note: This table reports the contribution of macro, debt issuance, and FOMC events as a share of the total post-Covid change in the 10-year yield. Each share multiplies the pass-through $\hat{\gamma}_k^h$ from Equation 4, estimated over 2013–2026, by the post-Covid sum of window moves of category k. ***, **, * denote significance at the 1%, 5%, and 10% levels based on heteroskedasticity-robust standard errors, respectively.

3.3 Average Event Shocks

With pass-through stable, the larger post-Covid contributions of macro and debt issuance events must come from the shocks. The contribution of each category can be written as $C^k = \gamma^k \times N^k \times \bar{\lambda}^k$, the product of pass-through γ^k, the number of events N^k, and the average window move per event $\bar{\lambda}^k$. The previous subsection shows that γ^k is stable across periods. The frequency of events is also stable, since data releases, Treasury announcements and auctions, and FOMC meetings follow similar schedules across years. The shift therefore lies in $\bar{\lambda}^k$.

The average shock increases after Covid for macro and debt issuance events (Table 4, columns 1–3). The average macro shock more than triples, from 0.05bp to 0.18bp per release, and the average debt issuance shock turns from −0.10bp to +0.06bp. The average FOMC shock remains negative but shrinks in magnitude, from −0.45bp to −0.16bp. Multiplying each average by the number of events recovers the contributions in Table 1.

To rule out that these shocks reflect intraday or calendar patterns rather than the event, we estimate them in a regression at the level of 15-minute intervals with seasonality controls. Specifically, we estimate

$$\Delta i_t = \alpha + \underbrace{\lambda^m \mathbb{1}_t^m}_{\text{macro}} + \underbrace{\lambda^d \mathbb{1}_t^d}_{\text{debt issuance}} + \underbrace{\lambda^f \mathbb{1}_t^f}_{\text{fomc}} + \Gamma \mathbb{X}_t + \epsilon_t, \tag{5}$$

Table 4: Average Shock By Event Category

	Simple average (bp per event)			Regression (bp per event)			H_1: post > pre
	Full	Pre	Post	Full	Pre	Post	(one-sided p)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Macro	0.101**	0.052	0.180*	0.121**	0.077	0.194*	0.161
	(0.049)	(0.048)	(0.103)	(0.051)	(0.050)	(0.108)	
Events	2,533	1,568	965	2,533	1,568	965	
Bars per event				3.00	3.00	3.00	
Prices	-0.013	0.181	-0.315	-0.019	0.188	-0.343	0.897
	(0.175)	(0.123)	(0.404)	(0.175)	(0.123)	(0.402)	
Real activity	0.145**	0.057	0.287**	0.155**	0.074	0.285**	0.081
	(0.066)	(0.068)	(0.133)	(0.066)	(0.068)	(0.134)	
Surveys	0.071	-0.013	0.212	0.125*	0.025	0.292*	0.060
	(0.072)	(0.072)	(0.149)	(0.075)	(0.076)	(0.155)	
Debt issuance	-0.039	-0.103**	0.060	-0.029	-0.098**	0.075	0.021
	(0.038)	(0.044)	(0.069)	(0.040)	(0.046)	(0.072)	
Events	1,310	792	518	1,310	792	518	
Bars per event				3.00	3.00	3.00	
FOMC	-0.329	-0.445	-0.164	-0.356	-0.483*	-0.168	0.233
	(0.217)	(0.293)	(0.323)	(0.217)	(0.291)	(0.319)	
Events	199	117	82	199	117	82	
Bars per event				2.58	2.63	2.50	
Observations (bars)				294,075	182,945	111,130	
R^2				0.0002	0.0004	0.0003	

Note: This table reports the average 10-year yield change in macro, debt issuance, and FOMC event windows, in basis points per event, for the full, pre, and post samples. Columns 1 to 3 report simple averages, columns 4 to 6 the coefficients from Equation 5, and column 7 the one-sided p-value for H_1: column 6 > column 5. Because Equation 5 is a regression at the level of 15-minute intervals, columns 4 to 6 multiply the estimated coefficients and standard errors by the average number of intervals per event in each category and period (the *intervals per event* rows), so that they read in bp per event and are comparable with columns 1 to 3. The FOMC statement window (e.g., 1:45pm to 2:30pm) and the press conference window (e.g., 2:15pm to 3:30pm) share one 15-minute interval, so FOMC events have slightly fewer than 3 intervals on average. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively, with t-test for columns 1 to 3, and trading-day-clustered standard errors for columns 4 to 6. Within R^2 is reported.

where Δi_t is the 15-minute change in the 10-year yield, $\mathbb{1}_t^k$ indicates that interval t lies in a window of category k, and the seasonality controls $\mathbb{X}_t$ contain hour-of-day and day-of-week fixed effects together with indicators for the first and last trading days of each month and quarter. For macro events, we also estimate a version that splits releases into prices, real activity, and surveys. Non-event intervals are the omitted baseline, so λ^k measures how much more the yield moves in an event interval than in a non-event interval at the same time of day and point in the calendar. We scale λ^k by the average number of intervals per event, so that it reads in basis points per event.

The pattern is robust to seasonality controls (Table 4, columns 4–6). The regression estimates are close to, and slightly larger than, the simple averages. The post-Covid increase is significant for debt issuance events, for real-activity and survey releases. For macro releases as a whole, the increase is not statistically significant, because price releases move the yields down on average after mid-2021.

The post-Covid increase in macro moves is concentrated in real-activity and survey releases (Figure 4; the full set is in Table A4). The employment report moved the 10-year yield up by 1.9bp per release after 2021, about four times its pre-Covid average; ISM non-manufacturing by 1.5bp, against 0.2bp before; retail sales and housing starts also turned more positive. Price releases, by contrast, moved yields down on average after 2021: PPI (−0.6bp) and unit labor costs (−1.2bp). Among FOMC events, statements turned from lowering yields (−0.5bp) to raising them (+0.3bp), while press conferences lowered them in both samples (−0.4bp and −0.7bp), consistent with statements and press conferences representing distinct phases of FOMC communication (Cieslak, McMahon and Pang, 2024). Among debt issuance events, refunding statements and bond auction results push yields up after 2021, while the negative drift after note auctions disappears.

3.4 Macro News or the Response to It?

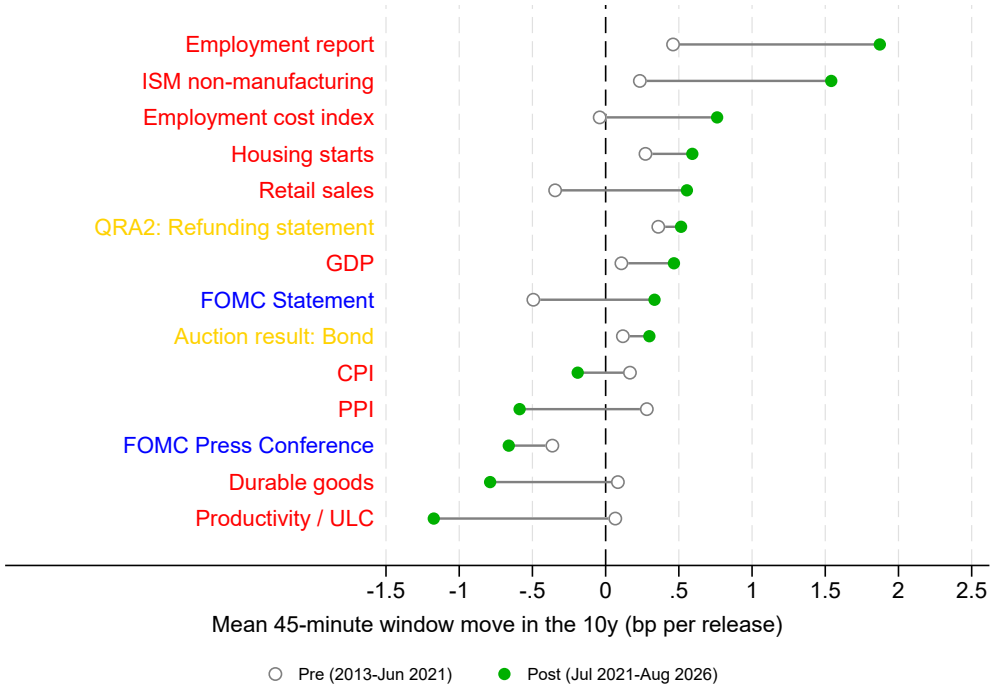
A larger average macro shock can arise in two ways. The macro news content may have changed, with surprises becoming larger or more one-sided. Alternatively, yields may have become more sensitive to a given unit of news. Bloomberg consensus forecast errors let us separate the two, at least for the headline forecast of each release.

The average size of macro news did not change much (Table 5). Price and real-activity news averaged 0.12 and 0.02 standard deviations post Covid, neither significantly different from before. Survey news turned negative, from 0.13 to −0.06, suggesting survey releases, on average, disappointed post Covid. The dispersion of news is also similar across samples, with the standard errors in comparable magnitude.

To measure the sensitivity to macro news, we regress each release's window move on its forecast error in the panel of releases, allowing the sensitivity to differ between samples:

$$\Delta i_{jt}^{45m} = \alpha_j^{\text{pre}} + \alpha_j^{\text{post}} + \beta^{\text{pre}} \mathbb{1}\{t \in \text{pre}\}\, e_{jt} + \beta^{\text{post}} \mathbb{1}\{t \in \text{post}\}\, e_{jt} + \Gamma \mathbb{X}_{jt} + \epsilon_{jt}, \tag{6}$$

Figure 4: Average Shocks By Detailed Events



Note: This figure plots the average 10-year yield change in macro, debt issuance, and FOMC event windows by detailed event type, in basis points, for the pre and post samples. Events are selected by the size of their average post-Covid move: top nine, bottom four, and CPI.

Table 5: Average Size of Macro News

	Prices			Real activity			Surveys		
	Pre	Post	p-value	Pre	Post	p-value	Pre	Post	p-value
Mean	0.003	0.119	0.406	−0.018	0.019	0.624	0.132	−0.062	0.017
Standard error	(0.081)	(0.113)		(0.045)	(0.061)		(0.051)	(0.063)	
N of events	201	135		884	545		474	285	

Note: This table reports the mean, its standard error (clustered by trading day), and the number of events for each group of macro news, i.e., Bloomberg forecast errors scaled by each release's standard deviation over January 2013–February 2020. The p-value tests H_0: the pre-Covid mean equals the post-Covid mean.

where Δi_{jt}^{45m} is the 45-minute change in the 10-year yield around release j on day t and e_{jt} is the scaled forecast error. The release-by-sample fixed effects α_j^{pre} and α_j^{post} absorb the average window move of each release, so β is identified from within-release, within-sample variation in surprises and reads in basis points per standard deviation of news. The regression includes the same seasonality controls as before. We estimate a pooled β for all releases and

group-specific β's for prices, real activity, and surveys.

Table 6: Sensitivity to Macro News

	Pooling		Allowing for groups		
	β^{pre}	β^{post}	β^{pre}	β^{post}	p-value: $\beta^{post} = \beta^{pre}$
All releases	0.381***	0.642***			0.008
	(0.041)	(0.090)			
Prices			0.432***	1.800***	<0.001
			(0.103)	(0.323)	
Real activity			0.316***	0.370***	0.627
			(0.050)	(0.099)	
Surveys			0.546***	0.715***	0.284
			(0.075)	(0.139)	
Observations	2,524		2,524		
R^2	0.102		0.129		
Seasonal controls	Yes		Yes		

Note: This table reports the sensitivity of the 45-minute 10-year yield change to macro news (Bloomberg forecast errors scaled by each release's standard deviation over January 2013–February 2020), estimated from Equation 6. Release-by-sample fixed effects and seasonality controls are included. The "Pooling" block allows β to differ between the pre- and post-Covid samples; the "Allowing for groups" block allows it to differ also across groups of releases. 9 of the 2,533 events have no Bloomberg outturn or forecast. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively. Standard errors are clustered at the trading-day level.

The sensitivity of the 10-year yield to macro news heightens after 2021, driven by price releases (Table 6). Pooling all releases, the sensitivity rises from 0.4bp to 0.6bp per standard deviation of news. The response to inflation news rises about fourfold, from 0.4bp to 1.8bp, while the responses to real-activity and survey news change little and insignificantly.

The monetary policy regime is a natural explanation for why inflation news became more potent. When inflation is far above target and the Fed is tightening, inflation data map more directly into the expected policy path (Bauer, Pflueger and Sunderam, 2024; Elenev et al., 2024). We split the sample into five regimes: the zero lower bound (ZLB) until the December 2015 liftoff, the normalization period until February 2020, the Covid ZLB from March 2020 until the March 2022 liftoff, the hiking and plateau phase until the first cut in September 2024, and the cutting phase thereafter. We then estimate

$$\Delta i_{jt}^{45m} = \sum_{r=1}^{5} \alpha_{jr} + \sum_{r=1}^{5} \sum_{g=1}^{3} \beta_{rg} \, \mathbb{1}\{r(t) = r, \; j \in g\} \, e_{jt} + \Gamma \mathbb{X}_{jt} + \epsilon_{jt}, \tag{7}$$

where r is the regime, g indexes the three groups of releases, and α_{jr} are release-by-regime fixed effects.

The jump in sensitivity to inflation news is specific to the hiking phase (Table 7). The response to inflation news is stable at 0.4–0.6bp per standard deviation across the three regimes before the March 2022 liftoff. It rises to 3.7bp during the hiking and plateau phase and falls back to 0.7bp once the Fed starts cutting. Real-activity and survey news show no comparable spike: their hiking-phase coefficients (0.6 and 0.8) lie within the range of the earlier regimes. Their sensitivity is lowest during the Covid ZLB, in line with the state-dependent attenuation of yield responses to macroeconomic news near the lower bound documented by Swanson and Williams (2014). The evidence points to state-dependent sensitivity: markets read inflation news through the lens of the Fed's response, and that lens magnified inflation data precisely when the Fed was fighting inflation.

Table 7: Sensitivity to Macro News By Regime

	ZLB 2013–15	Normalisation Dec 2015–Feb 2020	ZLB, COVID Mar 2020–Mar 2022	Hiking/plateau Mar 2022–Sep 2024	Easing from Sep 2024
Prices	0.423***	0.602***	0.537***	3.673***	0.729*
	(0.160)	(0.193)	(0.128)	(0.698)	(0.391)
N of events	67	102	54	64	49
Real activity	0.758***	0.362***	0.093	0.621***	0.401***
	(0.140)	(0.092)	(0.057)	(0.197)	(0.142)
N of events	309	437	213	276	194
Surveys	0.736***	0.777***	0.279***	0.784***	0.406*
	(0.137)	(0.116)	(0.103)	(0.226)	(0.212)
N of events	168	231	112	139	109
Observations			2,524		
R^2			0.195		

Note: This table reports the sensitivity of the 45-minute 10-year yield change to macro news (Bloomberg forecast errors scaled by each release's standard deviation over January 2013–February 2020), estimated from Equation 7. Release-by-regime fixed effects and seasonality controls are included. 9 of the 2,533 events have no Bloomberg outturn or forecast. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively. Standard errors are clustered at the trading-day level.

Sensitivity and contribution, however, are different objects. A release's contribution to the level of yields depends on its average move, while sensitivity is a slope; a large slope adds to the level of yields only if surprises lean materially sizable in one direction. CPI releases moved the 10-year yield a great deal in both directions, so their large slope left a small, even negative, net contribution. For instance, in September 2022, core CPI rose 0.6 percent against

a 0.3 percent consensus; the 10-year yield rose 12bp in the window, and the S&P 500 fell 4.3 percent on the day, its worst day since June 2020. However, in November 2022, core CPI rose 0.3 percent against a 0.5 percent consensus; the 10-year yield fell 16bp in the window and about 30bp on the day, and the S&P 500 rose 5.5 percent, its largest gain since 2020.

Combining Table 5 and Table 6 shows the limited contribution from the macro headline forecast errors. Since the average window move of a release equals its fixed effect plus $\beta\,\bar{e}_j$, the headline component of a group's contribution is its sensitivity times its mean surprise times its number of releases. After 2021, this component is about 20bp of the 174bp macro contribution. Before Covid it was about 29bp of 82bp. This is consistent with headline forecast errors potentially missing the information within releases, such as revisions and components underneath headline numbers (Gürkaynak, Kısacıkoğlu and Wright, 2020). For instance, for the employment report, Bloomberg forecast error is based on payroll, while the report also includes labor participation rate, sectoral distribution and historical revisions of the payrolls.

3.5 Expected Rates or Term Premium?

The previous subsections show when the 10-year yield was repriced, but not which component of the yield the events moved. A yield can rise because markets expect higher short rates over the life of the bond, or because they demand more compensation for bearing duration risk. Expected short rates reflect the anticipated stance of monetary policy, including the perceived long-run neutral rate. The term premium reflects compensation for interest-rate, inflation, and duration risk. This subsection asks how each category of shocks splits between the two components.

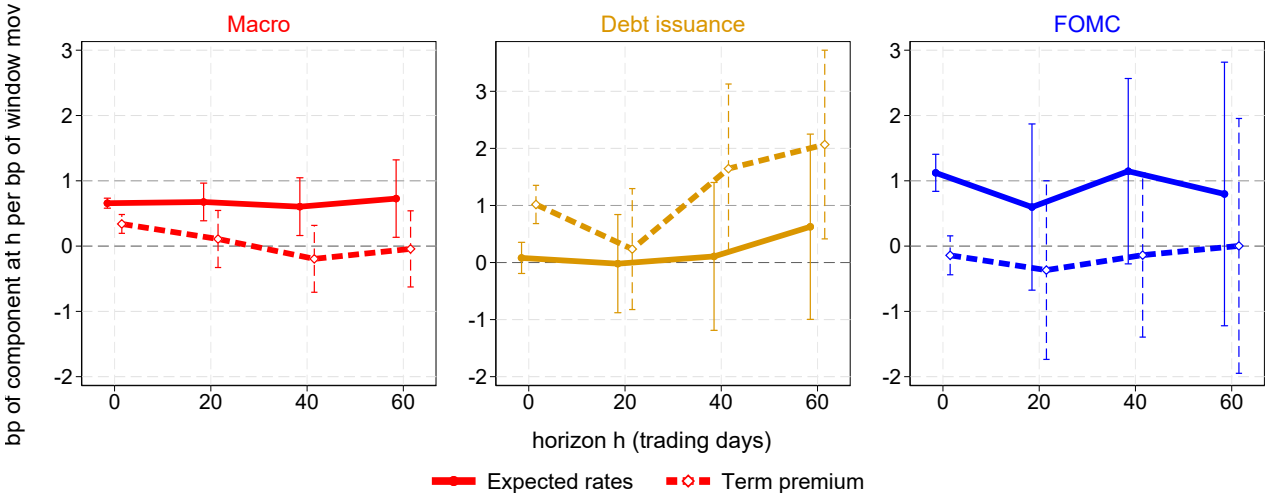
We estimate the response of the expected-rate and term-premium components of the 10-year yield (Adrian, Crump and Moench, 2013) to event shocks with local projections:

$$\Delta y_{t,t+h} = a + \underbrace{\beta^h_m m_t}_{\text{macro}} + \underbrace{\beta^h_d d_t}_{\text{debt issuance}} + \underbrace{\beta^h_f f_t}_{\text{fomc}} + \sum_{l=1}^{5} \rho_l\, \Delta i_{t-l} + \Gamma \mathbb{X}_t + \epsilon_{t+h}, \qquad y \in \{\text{expected rates}, \text{term premium}\}, \tag{8}$$

where $\Delta y_{t,t+h}$ is the change in the expected-rate or term-premium component from the close of day $t-1$ to day $t+h$, and m_t, d_t, and f_t are the sums of the 45-minute 10-year yield moves in macro, debt issuance, and FOMC windows on day t. The controls are five lags

of the daily change in the 10-year yield and seasonality indicators. Both components are regressed on the same controls, so the two responses sum to the response of the yield.

Figure 5: Impulse Responses of Event Shocks on Expected Rate and Term Premium



Note: This figure plots the impulse responses estimated from Equation 8 at horizons $h = 0, 20, 40, 60$ days for the post-Covid period. The whiskers are 90% confidence intervals, based on heteroskedasticity-robust standard errors.

Debt issuance shocks move the 10-year yield almost entirely through the term premium (Figure 5). On impact, debt issuance shock raises the term premium roughly one for one and leaves expected short rates essentially unchanged. This is the pattern predicted by models in which the price of duration risk depends on the quantity of duration that arbitrageurs must hold (Greenwood and Vayanos, 2014; Vayanos and Vila, 2021). Refunding announcements and coupon auctions reveal how much duration the private sector must absorb and at what price, but carry little news about the path of the policy rate. The result also matches accounts of the 2023 run-up in long-term yields, which coincided with the increase in coupon issuance announced at the August 2023 refunding and which model-based decompositions attribute mainly to the term premium (Diercks and Asnani, 2024).

FOMC and macro shocks, by contrast, move the 10-year yield mainly through expected short rates. On impact, FOMC shock raises the expected-rate component about one for one and the term premium hardly at all, the mirror image of debt issuance shocks. Monetary policy announcements thus work mainly by revising the expected path of the policy rate, consistent with the importance of the path factor in FOMC announcements (Gürkaynak,

Sack and Swanson, 2005*a*). Macro shocks load on the same component: a 1bp macro window move raises expected rates by 0.6bp and the term premium by 0.3bp on the event day. The expected-rate response stays near 0.7 at one to three months, whereas the term-premium response becomes insignificant within a month.

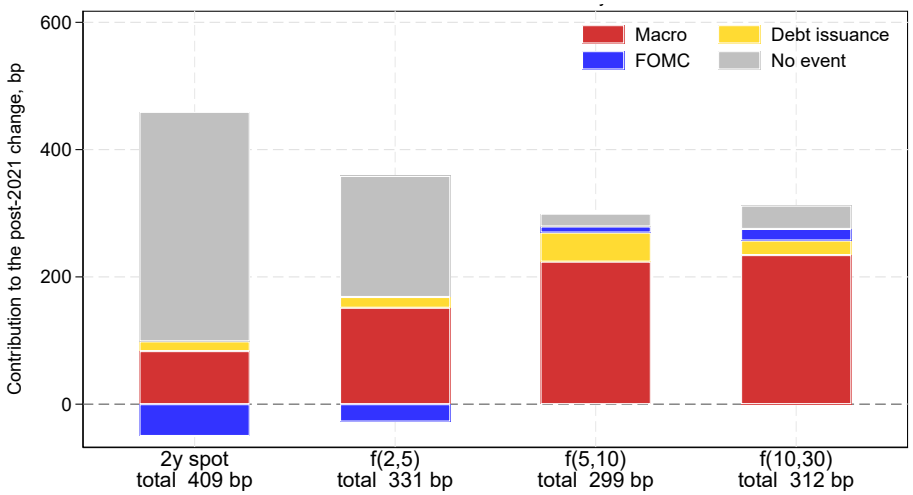
3.6 Yield Change Decomposition Across The Curve

The 10-year yield blends segments of the curve that price different information. The front end, e.g., the 2-year yield, is dominated by the expected policy path over the current cycle and should respond most to news about the Fed's near-term reaction. Forwards beyond five years lie largely beyond monetary policy horizon and mainly reflect the perceived long-run neutral rate and the term premium. Therefore, event categories could matter differently across these segments: FOMC announcements may carry news mostly about the near-term policy path, Treasury supply and demand news would matter most where duration risk is priced, and macro releases may move both. We repeat the accounting of subsection 3.1 for each curve segment, to explore their different roles across maturities.

The role of events rises steeply with maturity (Figure 6 and Table A3). The 2-year yield rose by about 410bp, of which macro and debt issuance windows explain a quarter and FOMC windows subtract about an eighth, while most of the increase accrued outside event windows. The pattern reverses beyond five years. Macro windows account for three quarters of the about 300bp rise in both $f(5,10)$ and $f(10,30)$, debt issuance windows add 15 and 7 percent, and non-event intervals contribute about one tenth or less. Macro news is the largest event contributor at every maturity. The contribution from macro and debt issuance events are statistically significant, and allowing for delayed effects leaves these shares nearly unchanged at the long end (Table A3).

Combined with subsection 3.5, the decomposition across the curve points to two channels at the long end. Macro news, which loads mainly on expected rates, raised long forwards by revising expected rates far beyond the current policy cycle. This is the excess sensitivity of long-term forward rates to economic news documented by Gürkaynak, Sack and Swanson (2005*b*), Beechey and Wright (2009), and Hanson, Lucca and Wright (2021), and it is consistent with markets revising their view of the long-run level of policy rates (Christensen and Rudebusch, 2026). Debt issuance news, which loads on the term premium, matters most

Figure 6: Contribution to Post-Covid Yield Changes By Event Category Across the Curve



Note: This figure decomposes post-Covid yield changes of different segments of the yield curve into cumulative changes in macro, debt issuance, and FOMC event windows and in non-event intervals (Equation 2), in basis points.

where duration risk is priced: its share is larger five years out, consistent with Treasury supply and demand operating through the term premium.

4 Second Moment Results: Bond-Stock Beta

This section turns from the changes in yields to the risk properties of Treasuries. The bond-stock beta represents whether Treasuries hedge equity risk, and hence the compensation investors require for holding Treasuries. This section aims to answer the question: where is the bond-stock comovement generated? The bond-stock beta is typically studied with daily or lower-frequency data, which cannot attribute comovement to specific events. Our event classification paired with intraday data allows us to decompose the beta into the same three categories of shocks, same as the first moment analysis in section 3.

4.1 Estimating Bond-Stock Beta and Its Sign Flip

We measure the bond-stock beta at the 15-minute frequency, following the conventional specification (e.g., Campbell, Pflueger and Viceira, 2026), and scale it per unit of duration so

that it is comparable along the curve:

$$r_t = \alpha + \beta \times s_t + \epsilon_t, \tag{9}$$

where $r_t = -100\,\Delta i_t$ is the duration-adjusted bond return, i_t is the yield of a given curve segment in basis points, and s_t is the return on the E-mini S&P 500 futures in basis points. β reads as the basis-point fall in the yield per 1 percent rise in the equity price, and equals 100 times the conventional beta divided by duration. A positive β means that bonds and stocks lose value together, so Treasuries are risky; a negative β means that Treasuries hedge equity risk. A conventional beta scales one for one with duration, so it rises with maturity even if every segment responds identically to equity news. Measuring returns per unit of duration removes this mechanical effect, and the result is invariant to the duration convention adopted for the forward segments.

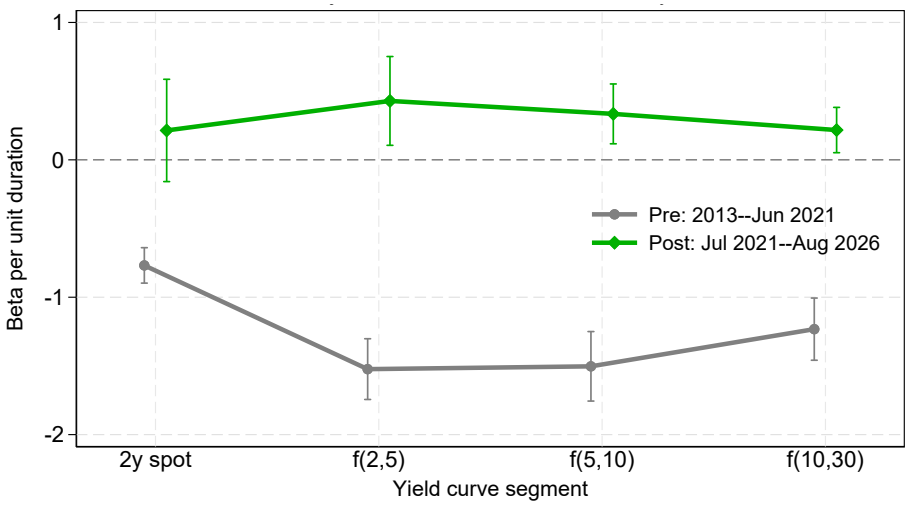
Intraday data confirm that Treasuries stopped hedging equities after 2021, and show that the flip occurs across the curve. The beta flips from negative to positive at every segment, and is significantly positive at the long end (Figure 7). Before mid-2021, Treasuries were a hedge: a 1 percent equity rally coincided with a 0.8bp rise in the 2-year yield and a 1.2–1.5bp rise in the forward rates. After 2021, the beta is significantly positive, between 0.2 and 0.4, for the forward rates. The shift is largest at the long end: about 1bp at the 2-year point against 1.5–2bp in the forward segments.

4.2 Decomposing the Beta by Event Category

Decomposing the beta by event category, an analogue of the first-moment accounting, would point to which shocks drive the sign flip. The bond return splits into $r_t = \sum_k r_t^k$, where r_t^k equals r_t in the 15-minute intervals of category k and zero otherwise. Because covariance is linear, the beta splits exactly into $\beta = \sum_k \beta^k$, where β^k is the slope of r_t^k on the full-sample stock return. The overall beta, to a close approximation (see Appendix B), is the weighted average of "within" beta of category:

$$\beta = \sum_k \beta^k \approx \sum_k b^k\, w^k, \tag{10}$$

Figure 7: Bond-Stock Beta Across the Yield Curve



Note: This figure plots the bond-stock beta estimated from Equation 9 across the yield curve, for pre- and post-period, respectively. β is defined as bp fall in yield per 1% stock rise. The whiskers represent 90% confidence intervals, with standard errors clustered at trading day level.

where b^k is the "within" beta of category k, estimated using only the intervals of that category, and w^k is the share of total stock-return variance that falls in those intervals. Intuitively, each category's covariance equals its within beta times the stock variance in its intervals, so each within beta enters the overall beta with weight w^k. The decomposition distinguishes two reasons why a category can matter for the overall beta: bonds and stocks may comove differently in its windows (a change in b^k), or its windows may carry more of the equity market's risk (a change in w^k).

Before Covid, Treasuries hedged equities at all times, not just around news (Table 8, Panel A). The within beta b^k is negative and of similar magnitude in macro, debt issuance, and non-event intervals, between -0.8 and -2.1 across maturities. Each category contributes roughly in proportion to its variance share, e.g., non-event intervals carry 92 percent of stock variance and account for a similar share of the overall beta. FOMC windows are the exception: their within beta is positive at the front end, consistent with monetary policy shocks, which move yields and stock prices in opposite directions (Jarociński and Karadi, 2020).

After Covid, the beta is generated inside event windows, with within betas turning positive

while comovement outside them largely disappeared. In macro windows, the within beta becomes strongly positive at every segment, from 4.3 at the 2-year point to 1.5 at $f(10, 30)$. Strong data now raise yields and lower stock prices, the "good news is bad news" pattern that arises when markets expect the Fed to lean against inflation and a strong economy (Boyd, Hu and Jagannathan, 2005; Elenev et al., 2024). In debt issuance windows, the within beta turns positive for forward yields, rising from 0.7 at $f(2, 5)$ to about 1.5 at the long end: news that raises the yield on duration now also lowers equity prices, plausibly through a common discount-rate effect. In non-event intervals, by contrast, the within beta collapses to essentially zero at the long end. FOMC windows keep their positive within beta, which grows to 6.3 at the 2-year point, consistent with announcement effects dominated by monetary policy news rather than central bank information (Jarociński and Karadi, 2020; Bauer and Swanson, 2023). At $f(5, 10)$, for instance, macro windows contribute 73 percent of the beta, debt issuance windows 15 percent, FOMC windows and non-event each below 10 percent.

The sign flip of beta comes from how bonds and stocks comove within event windows, not from those windows carrying more equity variance. Their shares of stock variance stay relatively stable, from 8 to 15 percent, and combining post-Covid weights with pre-Covid within betas yields an overall beta of -1.5 at $f(5, 10)$, essentially the pre-Covid value. The slightly larger weight of event windows amplifies the positive beta, but the dominant force is the change in within betas.

The composition of the beta changes along the curve in a way that mirrors the first moment. Macro windows account for most of the beta at the long end, from about 70 percent to above 90 percent for forward betas. The debt issuance contribution rises monotonically with maturity, from negative to 23 percent, for 2-year spot and $f(10, 30)$, respectively. FOMC windows matter most at the front end, accounting for 62 percent of the post-Covid beta at the 2-year spot, 19 percent at $f(2, 5)$, and almost none at $f(10, 30)$. Beyond five years, non-event intervals contribute little to either moment: about one tenth or less of the rise in long forward yields and of their betas.

These results locate the post-Covid change in the bond-stock beta more precisely than existing studies. The mechanism in Pflueger (2025) generates a positive beta when supply-driven inflation shocks interact with an inflation-focused Fed. Our evidence places this mechanism in macro releases, and adds a debt issuance channel at the long end. Duffee (2023) finds that macroeconomic news drives a substantial part of bond-stock comovement

Table 8: Bond-Stock Beta and Contribution By Event Category

		2y spot		$f(2,5)$		$f(5,10)$		$f(10,30)$	
	Var. share (%)	beta within	Contrib.	beta within	Contrib.	beta within	Contrib.	beta within	Contrib.
Panel A: Pre-COVID (Jan 2013–Jun 2021)									
Macro	4.7	-1.03***	-0.05	-2.07***	-0.10	-1.84***	-0.09	-1.52***	-0.07
		(0.22)		(0.36)		(0.33)		(0.29)	
Debt issuance	1.5	-0.82***	-0.01	-1.58***	-0.02	-1.61***	-0.02	-1.54***	-0.02
		(0.13)		(0.23)		(0.18)		(0.19)	
FOMC	1.8	1.32**	0.02	2.17**	0.04	0.92	0.02	0.47	0.01
		(0.56)		(0.95)		(0.72)		(0.72)	
Non-event	92.0	-0.80***	-0.73	-1.57***	-1.44	-1.53***	-1.41	-1.25***	-1.15
		(0.08)		(0.13)		(0.14)		(0.13)	
Total (overall beta)	100.0		-0.77***		-1.52***		-1.50***		-1.23***
			(0.08)		(0.13)		(0.15)		(0.14)
Panel B: Post-COVID (Jul 2021–Aug 2026)									
Macro	9.5	4.30***	0.41	4.36***	0.42	2.48***	0.24	1.54***	0.15
		(0.46)		(0.41)		(0.28)		(0.19)	
Debt issuance	3.3	-0.92	-0.03	0.66**	0.02	1.59***	0.05	1.54***	0.05
		(0.70)		(0.27)		(0.39)		(0.41)	
FOMC	2.0	6.30***	0.13	4.13***	0.08	1.27***	0.03	0.22	0.00
		(0.59)		(0.58)		(0.36)		(0.44)	
Non-event	85.2	-0.34*	-0.29	-0.11	-0.09	0.02	0.02	0.02	0.02
		(0.21)		(0.18)		(0.12)		(0.09)	
Total (overall beta)	100.0		0.21		0.43**		0.33**		0.22**
			(0.23)		(0.20)		(0.13)		(0.10)

Note: This table reports, for each event category, the share of stock-return variance (w^k), the within-category bond-stock beta (b^k), and the contribution to the overall beta (β^k) from the decomposition in Equation 10, for the pre-Covid (Panel A) and post-Covid (Panel B) samples. The "Total" row reports the overall beta from Equation 9. Bond-stock beta is scaled to the bp fall in the yield per 1% rise in S&P 500. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively, with standard errors clustered at trading day level.

at lower frequencies, but in ways that existing theories may not be able to fully match. At the 15-minute frequency, the link between events and the sign of the beta is direct. The two moments also tell a consistent story about the two channels. In macro windows, shocks load on expected rates and bonds fall with stocks, the pattern expected when markets price the Fed's reaction to a supply shock implied by the data. In debt issuance windows, shocks load on the term premium and bonds again fall with stocks, the pattern expected from a common shock to discount rates.

5 Conclusion

The rise in long-term Treasury yields after Covid occurs in places different from their movement before it. From 2013 to mid-2021, the movement in the 10-year yield occurs mostly outside the windows around key macro data releases, Treasury debt issuance announcements and auctions, and Federal Reserve monetary policy announcements. After mid-2021, macro and debt issuance windows, occupying 4 percent of trading time, account for over 60 percent of the rise in the 10-year yield, and their contribution increases with maturity.

The same windows account for most of the change in the riskiness of Treasuries. The bond-stock beta turns from negative to positive after Covid, and for long-term Treasuries the positive beta arises almost entirely in macro and debt issuance windows, while bond-stock comovement outside them mostly vanishes.

Together, the two sets of results point to two potential channels operating in these windows. In macro windows, data releases raise expected policy rates and lower stock prices, as markets price the Fed's reaction to the data. In debt issuance windows, news about Treasury supply and demand raises the term premium and lowers stock prices, consistent with a rise in risk premia common to bonds and stocks that is stronger at longer maturities.

These findings are relevant to both monetary policy and debt management. For monetary policy, they suggest that after Covid, long rates move less on FOMC announcements than on the market's reading of how the Fed will respond when macro data releases. For debt management, they suggest that Treasury debt issuance announcements and auctions matter not only for the level of long yields but also for the hedging value of Treasuries. One question remains open: our windows show which type of events move yields, not which structural shocks underlie the events. Separating demand, supply, monetary policy, and risk-premium shocks within event windows would test the two channels directly, and is the subject of a follow-up of this paper.

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A AI-Related Bond Issuance

Heavy issuance of long-dated bonds by AI-related companies has been cited as a driver of Treasury yields. The mechanism is that these corporate bonds compete with Treasuries for the same investors, so heavy supply of such long-dated corporate bonds could push Treasury yields up. If so, an AI-related bond announcement should move Treasury yields much like a Treasury issuance event does. We test this by measuring the change in the 10-year yield in 45-minute windows around the announcements, as in Equation 1.

Table A1: AI-Related Bond Deals

Deal	Announced	Amount ($bn)	Maturity (yrs)	Deal	Announced	Amount ($bn)	Maturity (yrs)
Oracle	30 Jan 2025	7.75	16.5	Amazon	10 Mar 2026	37.00	16.7
Alphabet	28 Apr 2025	5.00	24.3	Meta	30 Apr 2026	25.00	20.4
Oracle	24 Sep 2025	18.00	17.2	NVIDIA	15 Jun 2026	25.00	10.7
Meta	30 Oct 2025	30.00	19.3	SpaceX	23 Jun 2026	25.00	11.7
Alphabet	3 Nov 2025	17.50	20.5	Amazon	7 Jul 2026	25.00	14.6
Amazon	17 Nov 2025	15.00	15.7	Sopaipilla (Meta)	20 Jul 2026	12.30	22.3
Oracle	2 Feb 2026	25.00	16.2	Alphabet	6 Aug 2026	25.00	15.5
Alphabet	9 Feb 2026	20.00	15.3				
		Total issued (15 deals) / amount-weighted average maturity				312.55	16.5

Note: This table lists the issuance by announcement day, with the face value and its maturity. Maturity is the amount-weighted average, across the issuance's tranches, of the time from the issue date to the final maturity date (LSEG Deals; tranche proceeds as weights). The total row weights each tranche of the 15 deals by its proceeds.

We study U.S. dollar bonds issued by Alphabet, Amazon, Meta, NVIDIA, Oracle, SpaceX, and Sopaipilla (a special purpose vehicle for Meta's AI financing), between January 2025 and August 2026: 15 deals in total (Table A1).[3] Deal size and issue date come from LSEG Deals. We time-stamp each announcement by its first headline on IFR (LSEG News) and each pricing by the IFR pricing report, and assign both to the nearest 15-minute interval.[4] Announcements cluster around 8:00am ET and pricings around 4:00pm ET.

AI-related issuance rose from $93bn in 2025 to $219bn in the first eight months of 2026. Tranche maturities range from 2 to 50 years, with an amount-weighted average of 16.5 years.[5] In 2026, this issuance amounts to 8 percent of gross Treasury note and bond issuance by face

[3]About $87bn of issuance in other currencies by the same firms over the period is excluded.

[4]We checked for information released before the announcements. IFR reported SpaceX's deal on 12 June 2026, eleven days before its official announcement, and Oracle published a funding plan on Sunday, 1 February 2026. None of the earnings releases or calls we reviewed announced a bond.

[5]Maturities cluster at standard points: 2–3 years (18 tranches), 5 and 7 years (26), 10 years (14), 20 years (12), 30 years (14), 40 years (11), and 50 years (2).

Table A2: AI-Related Issuance Relative to Treasury Notes and Bonds

	Treasury (2y+)		AI-related		AI-related share (%)	
Period	Gross ($bn)	WAM (yrs)	Gross ($bn)	WAM (yrs)	Gross	Maturity-weighted
2025	4,223	7.4	93.25	18.6	2.2	5.5
2026 (Jan–Aug)	2,897	7.4	219.30	15.6	7.6	16.0
Total (Jan 2025–Aug 2026)	7,120	7.4	312.55	16.5	4.4	9.8

Note: Treasury is gross issuance (total accepted) of fixed-rate nominal notes and bonds with original maturity of two years or more (bills, TIPS, and FRNs excluded). AI-related bond issuance is the face value of the 15 deals in Table A1, by issue date; maturities are weighted by LSEG tranche proceeds within each deal. WAM is the amount-weighted average maturity at issue. The maturity-weighted share is AI-related issuance times WAM as a share of the same product for Treasuries.

value, and 16 percent weighted by maturity (Table A2).[6] Over January 2025-August 2026 as a whole, the shares are 4 and 10 percent. Relative to the Treasury market, AI-related issuance is small. The 15 deals total about 1 percent of U.S. GDP and of outstanding marketable Treasuries. Estimates relating long-term yields to projected federal debt imply about 2 to 3bp per percentage point of debt-to-GDP (Neveu and Schafer, 2024; Plante, Richter and Zubairy, 2026), while demand-system estimates imply about 13bp per percent of debt-to-GDP (Chaudhary, Fu and Zhou, 2024). Assuming AI-related corporate bonds were perfect substitutes for Treasuries, their issuance would raise long-term yields by between a few and about 14bp, a small contribution.

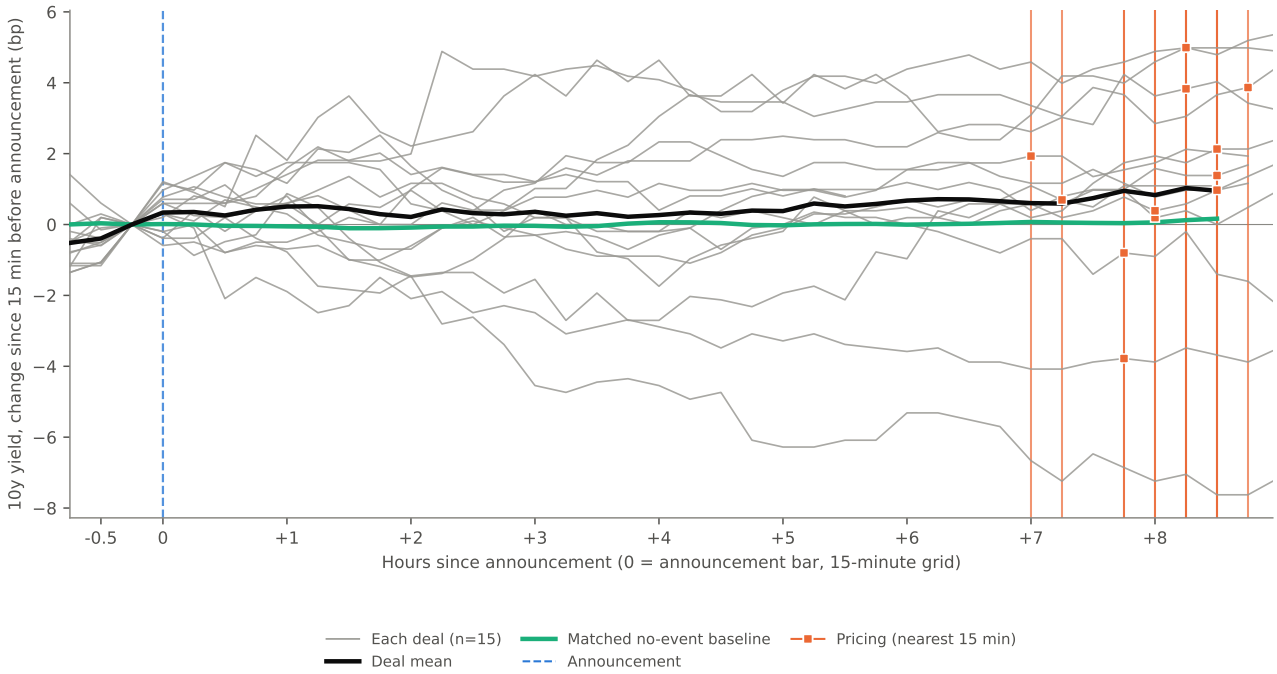
Treasury yields do not move significantly around AI-related announcements. The average announcement window is +0.25bp, with a standard error of 0.26bp. The absolute movements are also limited: their mean absolute move is 0.8bp, against 2.2bp for macro windows after Covid. Pricing windows are similarly small, averaging +0.2bp. Together from announcement to pricing, the 10-year yield rises by about 1bp more than over the same clock times on comparable non-event days, a difference within the range of ordinary variation (Figure A.1). One caveat remains, with only 15 events, these tests have limited statistical power.

B Details of Bond-Stock Beta Decomposition

Let s_t be the return on the E-mini S&P 500 futures in basis points, and $r_t = -100\,\Delta i_t$ the duration-adjusted bond return, where i_t is the yield in basis points. The 15-minute grid is

[6] Excluding the 40- and 50-year tranches, which exceed the longest Treasury maturity, the shares are 7 and 12 percent.

Figure A.1: 10-Year Yield Around AI-Related Bond Announcements



Note: This figure plots the 10-year yield change from the interval before the announcement. Orange lines mark pricing. The baseline is the mean of non-deal days with no macro, debt issuance, or FOMC event, over the same clock times.

partitioned into categories $k \in \{m, d, f, n\}$, macro, debt issuance, FOMC, and non-event. Let $\mathcal{B}^k$ be the set of 15-minute intervals in category k, N^k the number of intervals, and $p^k = N^k/N$ its share of total intervals over the full grid. Denote $\bar{r}^k$ and $\bar{s}^k$ for the averages of r_t and s_t over $\mathcal{B}^k$, and $\bar{s}$ for the full-sample average of s_t. The within-category moments are

$$\mathrm{Cov}^k(r,s) = \frac{1}{N^k}\sum_{t\in\mathcal{B}^k} r_t s_t - \bar{r}^k\bar{s}^k, \qquad \mathrm{Var}^k(s) = \frac{1}{N^k}\sum_{t\in\mathcal{B}^k} s_t^2 - \left(\bar{s}^k\right)^2.$$

The overall beta and the category component of the bond return are

$$\beta = \frac{\mathrm{Cov}(r,s)}{\mathrm{Var}(s)}, \qquad r_t^k = \begin{cases} r_t & t \in \mathcal{B}^k, \\ 0 & \text{otherwise.} \end{cases}$$

The contribution of category k is the slope of its component on the *full-sample* stock return:

$$\beta^k = \frac{\mathrm{Cov}(r^k, s)}{\mathrm{Var}(s)}$$

The within-category beta is the slope using *only the intervals in* $\mathcal{B}^k$:

$$b^k = \frac{\mathrm{Cov}^k(r, s)}{\mathrm{Var}^k(s)}$$

Step 1: additivity. Since $r_t = \sum_k r_t^k$ and covariance is linear,

$$\beta = \sum_k \beta^k$$

Step 2: expanding one category. Because r_t^k is zero outside $\mathcal{B}^k$, its mean over all N intervals is $p^k \bar{r}^k$, and

$$\begin{aligned}
\mathrm{Cov}(r^k, s) &= \frac{1}{N} \sum_{t \in \mathcal{B}^k} r_t s_t - p^k \bar{r}^k \, \bar{s} \\
&= p^k \Big[\mathrm{Cov}^k(r, s) + \bar{r}^k \bar{s}^k \Big] - p^k \bar{r}^k \, \bar{s} \\
&= p^k \, \mathrm{Cov}^k(r, s) + p^k \bar{r}^k \Big(\bar{s}^k - \bar{s} \Big).
\end{aligned}$$

Step 3: the factorisation. Substituting $\mathrm{Cov}^k(r, s) = b^k \, \mathrm{Var}^k(s)$ and dividing by $\mathrm{Var}(s)$,

$$\beta^k = b^k \, w^k + \eta^k, \qquad w^k = \frac{p^k \, \mathrm{Var}^k(s)}{\mathrm{Var}(s)}, \qquad \eta^k = \frac{p^k \, \bar{r}^k \Big(\bar{s}^k - \bar{s} \Big)}{\mathrm{Var}(s)}. \tag{11}$$

Here w^k is the share of stock-return variance in category k.

Step 4: the remainder is negligible. The remainder η^k is proportional to the product of the category averages $\bar{r}^k$ and $\bar{s}^k - \bar{s}$. At the 15-minute frequency these are tiny relative to the standard deviations of r_t and s_t, so $\eta^k \approx 0$. By the law of total variance,

$$\sum_k w^k = 1 - \frac{\sum_k p^k \Big(\bar{s}^k - \bar{s} \Big)^2}{\mathrm{Var}(s)} \approx 1$$

In estimation we use the uncentred share $w^k = \sum_{t\in\mathcal{B}^k} s_t^2 \Big/ \sum_{t=1}^{N} s_t^2$, which coincides with the expression in Equation 11 when the averages are zero and sums to one exactly. Therefore

$$\beta = \sum_k \beta^k \;\approx\; \sum_k b^k\, w^k \tag{12}$$

that is, the beta is a weighted average of the within-category betas, with weights equal to the categories' shares of stock-return variance.

C Additional Tables

Table A3: Contribution to Yield Changes By Event Category Across Yield Curve

		Macro		Debt issuance		FOMC		Non-event	
Segment	Δ (bp)	Raw	$h = 22$	Raw	$h = 22$	Raw	$h = 22$	Raw	$h = 22$
2y spot	409.3	20.4	14.8***	3.7	5.4**	-12.2	-16.0**	88.1*	95.8
$f(2, 5)$	331.3	45.7	28.5***	5.1	3.5	-8.3	-10.4***	57.4	78.3**
$f(5, 10)$	299.3	74.8***	71.4***	15.3	16.8***	3.2	4.3*	6.7	7.6***
$f(10, 30)$	312.0	75.1***	72.3***	7.3	7.6**	5.8	8.8*	11.7	11.3***

Note: This table reports the cumulative yield changes in the macro, debt issuance, and FOMC event windows as share of the total change, for post-Covid period. Total yield change in basis points is reported in column "Δ bp". Simple accounting is reported under column "Raw," regression-based (Equation 4) accounting for $h = 22$ days is reported under column "$h = 22$." ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively, with t-test for column "Raw," and heteroskedasticity-robust standard errors for column "$h = 22$."

Table A4: Average Shock By Detailed Event, Full Set

	Full		Pre		Post	
Release	Mean	N	Mean	N	Mean	N
FOMC						
FOMC Press Conference	-0.507	85	-0.364	44	-0.661	41
FOMC Statement	-0.196	114	-0.494	73	0.334	41
Total events		199		117		82
Debt issuance						
QRA1: Borrowing estimates	0.163	55	0.174	34	0.145	21
QRA2: Refunding statement	0.419*	55	0.360	34	0.515	21
Auction announce: Bond	0.154	26	0.076	5	0.172	21
Auction announce: Note 10y	0.000	110	0.044	69	-0.073	41
Auction announce: Note other	-0.003	164	-0.003	102	-0.003	62
Auction result: Bond	0.212	81	0.118	39	0.299	42
Auction result: Note 10y	-0.160	166	-0.291	104	0.059	62
Auction result: Note other	-0.117**	653	-0.191***	405	0.002	248
Total events		1310		792		518
Macro						
Housing starts	0.401***	144	0.272*	86	0.592**	58
CPI	0.032	160	0.166	100	-0.191	60
Conference Board consumer confidence	-0.081	164	-0.114	102	-0.026	62
Durable goods	-0.247	158	0.084	98	-0.789*	60
Employment cost index	0.334	30	-0.041	16	0.762	14
Employment report	0.982**	157	0.461	99	1.873**	58
GDP	0.242	150	0.108	94	0.467	56
ISM non-manufacturing	0.705***	111	0.234	71	1.541***	40
ISM manufacturing	-0.100	164	-0.023	102	-0.226	62
Jobless claims	0.220	163	0.183	102	0.282	61
Leading index	0.049	148	-0.037	92	0.191	56
New home sales	-0.298*	115	-0.285*	75	-0.322	40
Productivity / ULC	-0.334	31	0.066	21	-1.174	10
PPI	-0.076	124	0.282*	73	-0.587*	51
Industrial production	-0.090	162	-0.134	102	-0.015	60
Retail sales	0.012	149	-0.345	90	0.555	59
Trade balance	0.016	83	0.302	46	-0.339	37
U. Michigan consumer confidence	0.017	320	-0.045	199	0.119	121
Total events		2533		1568		965

Note: This table reports the average cumulative 10-year yield changes in each macro, debt issuance, and FOMC event window in basis points, for pre and post samples, respectively. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively, with standard errors clustered at trading day level. Within macro events, windows are assigned exclusively based on the percentage of Bloomberg users who set an alert for a particular event. See section 2 for details.



PUBLICATIONS

High-Frequency Evidence on Post-Covid Treasury Yields and Bond-Stock Comovement

Working Paper No. WP/2026/247